# Spatial Patterns of NDVI Vegetation Greenness Index and NDWI Water Content Index Using Sentinel 2A Imagery

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# Spatial Patterns of NDVI Vegetation Greenness Index and NDWI Water Content Index Using Sentinel 2A Imagery

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**Abstract.** Sentinel-2A imagery in forestry and environmental studies is critical to keep up with the changes in dynamic forest ecosystems. Sentinel-2A is a European optical imaging satellite forester that supports forest monitoring services, land cover change detection, and disaster management. The study aimed to analyze the vegetation greenness index with the NDVI [Normalized Difference Vegetation Index] and the NDWI water content index [Normalized Difference Water Index] on various land covers. The research method used was digital transformation techniques of both the NDVI and NDWI indices. The results showed that processing image classification in a combination of NDVI and NDWI index values resulted in various land cover variations with different accuracies. The advantages and disadvantages of the variety of index values suggest that it facilitates interpretation in classifying land cover classes. However, distinguishing vegetation classes in terms of color is difficult because some objects' coloring is almost the same.

## 1. Introduction

Sentinel-2 is a satellite launched in collaboration between The European Commission and the European Space Agency in the Global Monitoring for Environment and Security (GMES) program. Sentinel-2 was designed to ensure the continuation of Landsat 5/7, SPOT-5, SPOT-Vegetation, and Envisat MERIS missions that will end their operations. The Sentinel-2 mission confirms Europe's commitment to assisting the world in earth observation activities, which continues by using several instruments with different spatial and spectral resolutions, faster temporal resolutions, and globally covered areas. Sentinel-2 is specifically designed to help scientists study and monitor earth interactions and processes, prepare strategies in the face of the challenges of global change, and achieve community development (Societal Development Goals). In addition, the satellite expects scientists to get new data harmonization and scientific products to develop and integrate direct monitoring through modeling to understand regional and global environmental changes and find solutions to anticipate and manage destructive climate change [1–3].

Sentinel-2 satellite channel specification refers to the canals contained in SPOT and Landsat. Changes in the canal width and the addition of the canal were made to Sentinel-2 to improve performance in earth observation—spectrum coverage of each channel and spatial resolution of Sentinel 2, SPOT, and Landsat. Thirteen canals installed on the Sentinel-2 satellite have their characteristics. Four canals with a spatial resolution of 10 m ensure compliance with SPOT 4/5 and meet user requirements for land cover classification. The spatial resolution of 20 m possessed by six channels becomes required for other levels of two processing parameters. Canals with a spatial resolution of 60 m are devoted to atmospheric correction and cloud filtration (443 nm for aerosols, 940 nm for water vapor, and 1375 for thin cloud detection). A resolution of 60 m is considered sufficient to capture the spatial variability of atmospheric and geophysical parameters [4–6].

Sentinel-2 leverages the technology and experience gained in Europe and the United States. Sentinel-2 supports the supply of operational data for services such as risk management (floods and forest fires, forest subsidence and landslides), land use/change, forest monitoring, food security/early warning systems, water management and land protection, urban mapping, natural hazards, terrestrial mapping for humanitarian and development assistance, and monitoring of land and seawater conditions. The Sentinel-2 satellite imagery has been used in the marine field and is very promising. However, the information on Sentinel-2 imagery for the maritime area, especially in Indonesia, is relatively new and can be accessed and downloaded for free [8–10].

The utilization of Sentinel-2 imagery in various fields of research has increased. Using sentinel-2 imagery in agriculture [11–13]; Dynamic changes in canopy water cane using Sentinel-2A and Landsat imagery [14]. Monitoring shoreline changes [14] and using Sentinel and Landsat data to analyze soil moisture [15]. Automatic water content detection [16–19] and spectral index exploitation on forest classification models in Morocco [20]. Extraction of river bodies using multi-scale area development methods [21], damage assessment of thrips chili [22], and comparative evaluation of the effect of atmospheric correction on radiometric indices [23]; Water level monitoring and river water quality parameters are based on passive and active multi-temporal sensing 2016-2020 [24], monitoring of lake changes [25], the computer model of tsunami vulnerability using sentinel data 2-A and SRTM [26], and identifying surface water resources [27]. Applications of sentinel imagery utilization for greenness research and water content on various land cover variations for small island areas are still relatively limited. Therefore, this paper will explore analyzing the spatial sample of NDVI greenery and NDWI water content material on several variants of land cover in Hutumuri Village of Ambon City.

## 2. Material and Methods

### *2.1. Research Area*

The research site selection is in Hutumuri Village, Ambon City, with an area of ± 2,360 hectares. The research area was determined by considering the relatively high vegetation canopy stratification, according to the Map of Forest Area and Water Conservation of Maluku Province [Annex to Decree of the Minister of Forestry Number SK. 854/Menhut-II/2014 dated September 29, 2014]. The research area includes forest-protected forests and other land uses. The topographic condition of Ambon City is that it is located on a small island, and the river in Ambon has a unique character composed of many small rivers with narrow watersheds. According to the river map, Hutumuri Village has rivers that flow throughout the year, known as perennial rivers.

### *2.2. Research Procedure*

The initial stage of sentinel image processing in NDVI and NDWI transformations is to perform radiometric and geometric corrections. Radiometric correction is a repair stage in the form of defects or errors due to disruption of electromagnetic radiation energy in the atmosphere and mistakes due to the influence of the elevation angle on the sun. The radiometric correction aims to correct pixel values to match what they should be. Errors in the optical system cause blurry images and changes in signal strength. Other errors result from interference with the energy of electromagnetic radiation in the atmosphere, giving rise to scattering and uptake, non-linear amplitude responses, and noise at the time of data transmission [23].

The geometric correction stage aims to regulate the Earth's coordinate system's function and existence so that all image data information corresponds to its presence on Earth. Geometry correction has three stages: (1) The image rectification (correction) or restoration corrects the coordinates of the image according to the coordinates of the geography. (2) The registration (matching) stage adjusts the image's position with other photos or transforms the coordinate system of a multispectral or multitemporal image. (3) We register an image to a map or transform an imagery coordinate system to a map to produce a photo with a particular projection system. Geometric correction ensures that differences in position between Sentinel images are not caused by geometric errors of imagery and other variations but rather due to limited water index capabilities. [28].

Sentinel image processing uses ENVI 5.3 software to obtain the vegetation index, the Normalized Difference Vegetation Index (NDVI), and the water index transformation, the Normalized Difference Water Index (NDWI). The water index (NDWI) is a band reflectance transformation for extracting greenness levels and water content, which includes two algorithms with mathematical equations for NDVI and NDWI [29], [30].

**Table 1.** Use of bands, wavelength, and spatial resolution in NDWI using Sentinel

| Band | Wavelenghth (μm) | Spatial resolution (m) |
|---|---|---|
| B1 Coastal aerosol | 0.433-0,453 | 60 |
| B2 Blue | 0,458-0,523 | 10 |
| B3 Green | 0,543-0,578 | 10 |
| B4 Red | 0,650-0,680 | 10 |
| B5 Vegetation Red Edge | 0,698-0,713 | 20 |
| B6 Vegetation Red Edge | 0,733-0,748 | 20 |
| B7 Vegetation Red Edge | 0,773-0,793 | 20 |
| B8 NIR | 0,785-0,900 | 10 |
| B8A Narrow NIR | 0,855-875 | 20 |
| N9 Water vapour | 0,935-0,955 | 60 |
| B10 SWIR-Cirrus | 1,365-1,385 | 60 |
| B10 SWIR-1 | 1,565-1,655 | 20 |
| B11 SWIR-2 | 2,100-2,280 | 20 |

**Source**: Sentinel-1A and 2A imagery combination [31]

NDVI (Normalized Difference Vegetation Index) is an image calculation used to determine the optimal greenness level at the beginning of vegetation classification. NDVI can indicate parameters related to vegetation, among others, green leaf biomass and green foliage areas, which are predictable values for vegetation division [12], [19], [32]. The mathematical equations of NDVI are:

$$NDVI = \frac{(\text{Band NIR} - \text{Band Red})}{(\text{Band NIR} + \text{Band Red})} \quad \text{...........} (1)$$

Where:
NDVI = Normalized Difference Vegetation Index (vegetation area index value)
NIR band = Near Infrared band (near-infrared channel) in the radiometrically corrected image
Red band = red channel in the radiometrically corrected image

NDWI Index separates water and non-water features. Several NDWI formulas combine different pairs of Landsat TM or ETM+ bands. NDVI uses two NIR channels, centered around 0.86 μm and about 1.24 μm [33–35].

$$NDWI = \frac{Band\ NIR - Band\ SWIR}{Band\ NIR + Band\ SWIR} \quad \text{...........} (2)$$

Where:
NDWI = Normalized Difference Water Index
NIR band = Near Infrared band (near-infrared channel) in the radiometrically corrected image
SWIR ban = infrared shortwave band (shortwave infrared channel) in the corrected image

## 3. Results and Discussion

### *3.1. Spatial pattern of NDVI Vegetation Index Period 2015-2020*

The calculation of NDVI vegetation greenness in 2015 showed that the greenery of NDVI, according to the highest to the lowest area, was the high greenness class (2262.46 Ha). There is a moderate greenness class (52.57 ha), a low greenness class (35.80 Ha), and a shallow greenness class (9.16 ha), as presented in Table 2 and Figure 1.

**Table 2**. NDVI Greenness level in the 2015 period (in hectares)

| Greenness class level | NDVI Class | Area (Hectares) |
|---|---|---|
| Very Low Greenness | -0,0089-0,1186 | 9,16 |
| Low Greenness | 0,1186-0,3257 | 35,80 |
| Medium Greenness | 0,3257-0,5328 | 52,57 |
| High Greenness | 0,5328-0,7399 | 2262,46 |

**Source**: Sentinel-2A imagery analysis of 2015

Spatial patterns of vegetation greenery in the study area during the 2015 period showed that the high greenness class dominated 95.87% of land closures, followed by average greenness at 2.23%, low greenness at 1.52, and very low greenness at 0.39%. Quiet and low-leaf greenery distribution patterns spread along with coastal-dominated settlements and dryland agriculture. High greenery dominates all land areas up to the mountains in the research area, as presented in Figure 2.

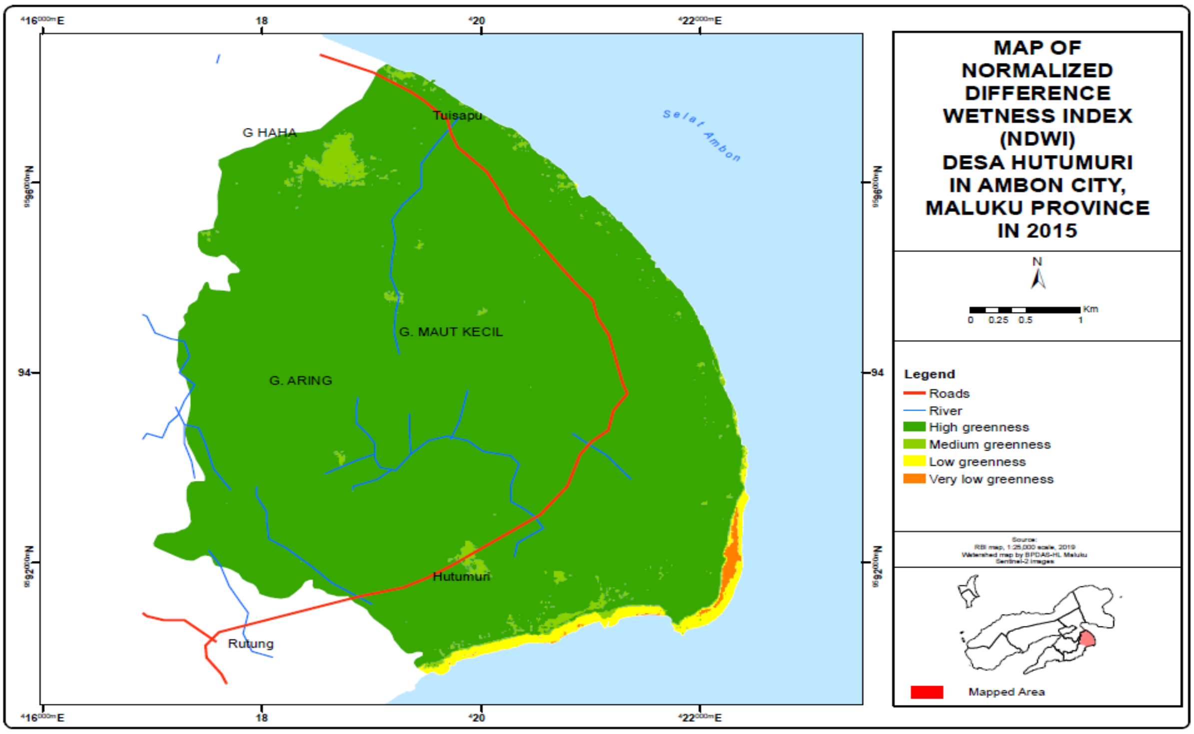


**Figure 1** Spatial pattern of vegetation index NDVI Hutumuri City Ambon City in 2015

The results of the calculation of greenery of NDVI vegetation in 2020 have changed compared to the 2015 period. NDVI greenness class with the highest to the lowest area is high greenness (2152.73 Ha); It is followed by low greenery (168.04 ha), very low greenery (33.27 Ha), and moderate greenery [5.96 ha] as presented in Table 3 and Figure 2.

**Table 3**. NDVI Greenness class in 2015 period (in hectares)

| Greenness class | NDVI Class | Area (Hectares) |
|---|---|---|
| Very Low Greenness | -0,0089-0,1186 | 33.27 |
| Low Greenness | 0,1186-0,3257 | 168.04 |
| Medium Greenness | 0,3257-0,5328 | 5.96 |
| High Greenness | 0,5328-0,7399 | 2152.73 |

**Source**: Sentinel-2A imagery analysis of 2020

Calculations of the greenness of NDVI vegetation in both periods showed changes in the high greenness class decreased by 4.65%, while the shallow greenness class increased by 6.73%, the low class decreased by 0.11%, and the greenness class decreased by 1.97%. The greenness of vegetation changed between the four classes of NDVI in the study period. This change is the impact of changes in land cover in research areas, especially the change of forests into non-forest regions, significantly increasing the number of settlements in Hutumuri Village, which continues to increase significantly.

The spatial pattern of greenness vegetation of NDVI in 2020 showed a significant change in the greenness class of shallow vegetation spreading in almost all areas of the study. Population growth and increased board or housing needs led to substantial changes in land closures. In contrast, the difference in the greenness class of low vegetation is still spreading along the coast and the 2015 period. The evolution of forest land cover to agricultural and settlement land occurred during the research period, as presented in Figure 3.

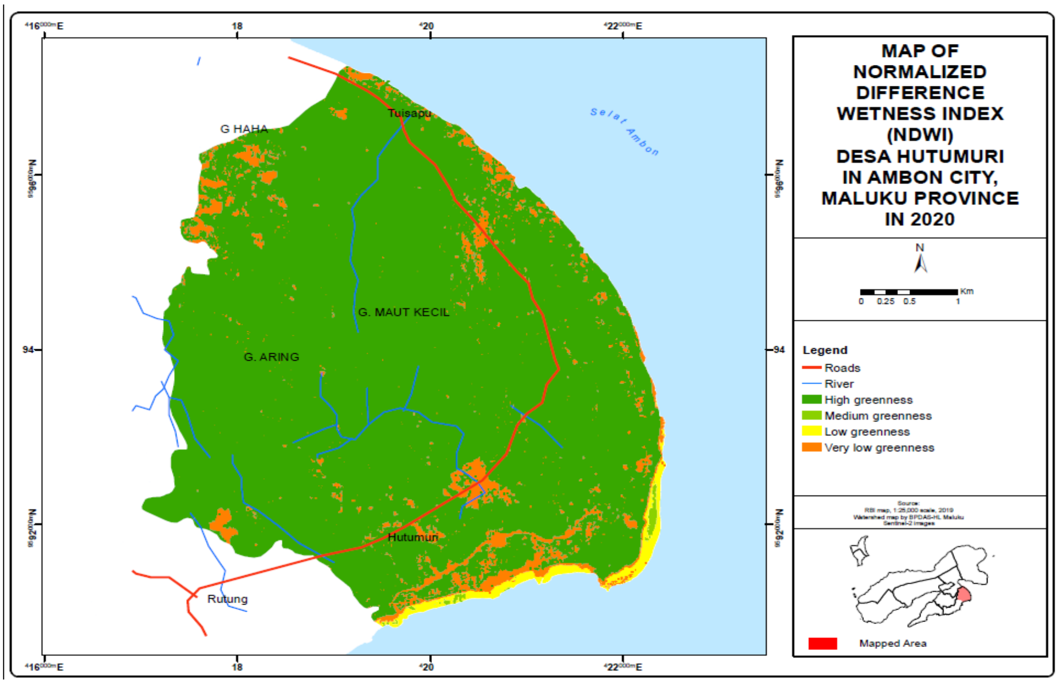


**Figure 2**. Spatial pattern of vegetation index NDVI Hutumuri City Ambon City in 2020

**Spatial pattern of NDWI water content for the period 2015-2020**

The results of the calculation of NDWI water content for the period 2015 showed that the highest area of NDWI water content was a medium water content of 1790.66 hectares, followed by a high water content of 295.10 hectares and a low water content of 274.24 hectares, as presented in table 4 and Figure 4.

**TABLE 4**. NDWI water content level in the period 2015 (in hectares)

| Water Content Level | Class of NDWI | Area (Hectares) |
|---|---|---|
| Low water index | -0.9497 s.d -0.5812 | 295.10 |
| Medium water index | -0.5812 s.d -0.2128 | 1790.66 |
| High water index | -0.2128 s.d 0.1556 | 274.24 |

**Source**: Sentinel-2A image analysis 2015

Spatial patterns are a vivid picture of a region on the Earth's surface. The NDWI water content map of Hutumuri State shows that the difference between the wet and surrounding dry areas is visible. Moderate water content (70.95%) dominates in Hutumuri Village, which is still dominated by tree vegetation. High water content spreads along the coast or other puddles. The last low water content (15.17%) and (13.88%) spread across the land to the hills and mountains. The spatial pattern of water distribution of NDWI Hutumuri State is presented in Figure 5.

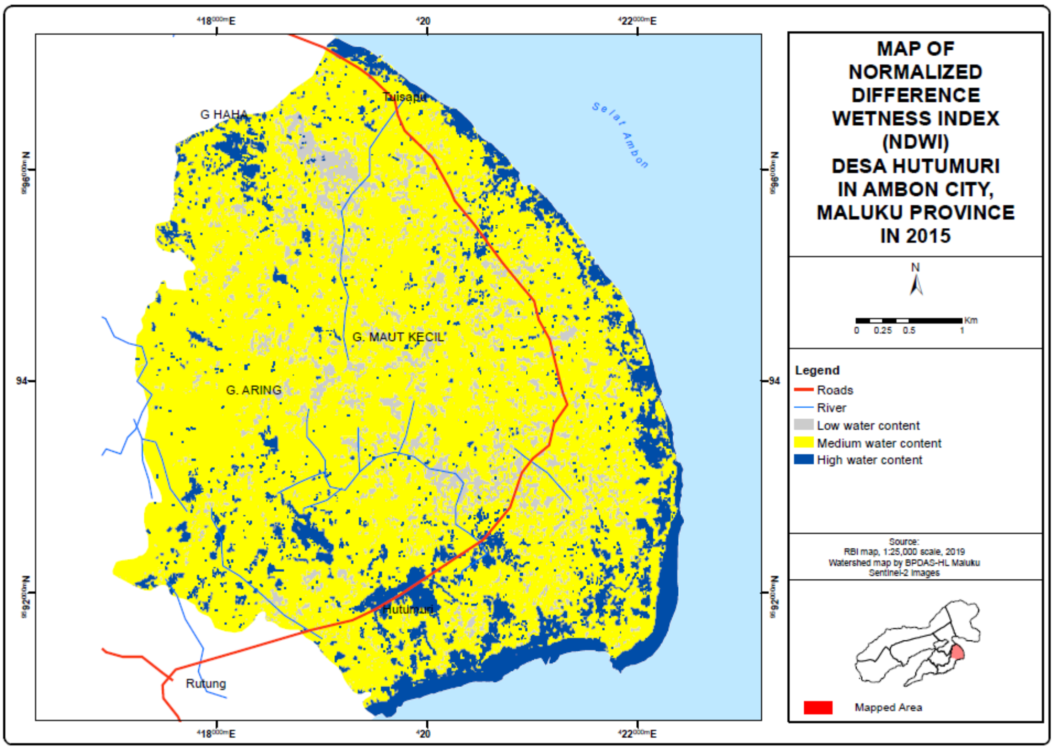


**Figure 3**. Spatial pattern of water content index NDWI Hutumuri Country Period 2015

The calculation results of the NDWI index for the period 2020 showed that the NDWI index with the highest area was a medium class of 1674.43 hectares, followed by the low class of 357.96 hectares and high class of 327.61 hectares, as presented in Table 5 and Figure 4.

**Table 5**. NDWI index level in the period 2020 (in hectares)

| Water Content Level | NDWI Class | Area (Hectares) |
|---|---|---|
| Low water index | -0.9497 s.d -0.5812 | 327.61 |
| Medium water index | -0.5812 s.d -0.2128 | 1674.43 |
| High water index | -0.2128 s.d 0.1556 | 357.96 |

**Source**: Sentinel-2A image analysis 2020

The pattern of distribution of water content of NDWI Hutumuri State in the period 2020 shows that changes in the distribution of NDWI water content are experiencing a pattern shift that extends to coastal areas towards the mainland. The difference between the wet area and the surrounding dry area is visible. Moderate water content still dominates, with the water content of tree vegetation beginning to experience decreased changes. In contrast, low and high water content increases and spreads along the coast, moving towards the land area. Spatial distribution patterns of low and high NDWI water content are applied from coastal and land areas to hills and mountains. The spatial pattern of water distribution of NDWI Hutumuri State period of 2020 is presented in Figure 4.

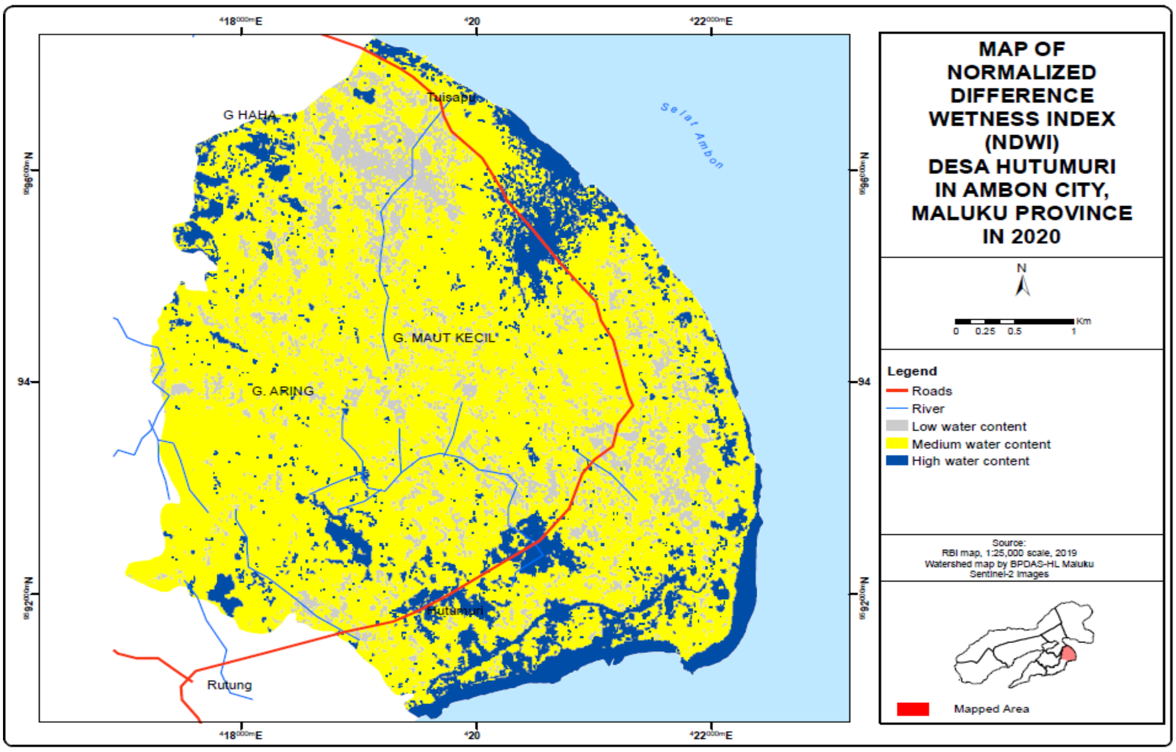


**Figure 4**. Spatial pattern of water content index NDWI Hutumuri Country Period 2020

Calculations of NDWI water content for 2015-2020 showed that the water content class change decreased by 4.92%, while the low NDWI water landing class increased (3.55%) and the high NDWI class increased (1.38%), as presented in Figure 4. The land cover classification results using sentinel-2A imagery analysis and the calculation of NDVI and NDWI index values showed that different land cover appeared. This is seen from the extent and accuracy resulting from the processing of this image classification. The area in each land cover class is relatively accurate, supported by the characteristics of each object quite clearly and accurately.

## 4. Conclusion

The transformation of the vegetation greenness index and the NDWI water index in the research area during the research period showed significant spatial pattern changes. During the study period, the spatial pattern of greenness vegetation of NDVI experienced a vital difference: the greenness class of vegetation was shallow spread in almost all areas of the study, while the change in the greenness class of low vegetation spread along the coast. During the study period, spatial patterns of NDWI water content showed that changes in NDWI water content experienced spatial pattern changes that extended to coastal areas towards land. The NDWI class is decreasing, followed by low and high NDWI classes increasing and spreading along the coast, moving towards the land area extending from coastal and land areas to hills and mountains.

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